GG-T-585B
November 19, 1986
SUPERSEDING
Int. GG-T-00585a (DSA-DM)
March 14, 1969
GG-T-585
November 4, 1953

FEDERAL SPECIFICATION

TONOMETER, OPHTHALMIC, SCHIOTZ

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATIONS

1.1 Scope. This specification covers mechanical tonometers suitable for measuring intraocular pressure.

1.2 Classification.

1.2.1 Grade and Type. Mechanical tonometers furnished under this specification shall be of one grade and type of Schiotz design.

2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. The following specifications and standards of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
QQ-A-250/4	Aluminum Alloy 2024, Place and Sheet.
QQ-A-250/8	Aluminum Alloy 5052, Place and Sheet.
QQ-A-250/11	Aluminum Alloy 6051, Place and Sheet.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Set-up, Paperboard.

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STANDARDS

Federal

FED-STD-123 Marking of Domestic Shipment (civilian agencies).

Military

MIL-STD-129 Marking for Shipment and Storage.
MIL-STD-147 Palletized Unit Leads.
MIL-STD-794 Parts and Equipment, Procedures for Packaging of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which include cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C. Atlanta, Chicago, Kansas City, MO., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

* Specification of the Committee on Standardization of Tonometers of the American Academy of Ophthalmology and Otolaryngology. (Specification No. 5 or latest edition)

(Copies may be obtained from American Academy of Ophthalmology, San Francisco, CA.)

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3. REQUIREMENTS

3.1 Certification.

* 3.1.1 The contractor shall certify that the product offered meets the salient characteristics described herein and conforms to the producers' own drawings, specifications, standards, and quality assurance practices, and is the same as the product sold in the commercial market place. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

* 3.1.2 Each instrument shall be accompanied with a certificate of inspection and test certifying that the instrument is dimensionally correct and has been calibrated. The certifiying laboratory shall be an independent establishment as approved by the contracting agency.

3.2 Material. The material used in the construction of this unit shall be of the highest quality and suitable for the purpose intended.

3.2.1 Frame. The frame shall be fabricated of 2024 aluminum alloy, conforming to temper T4 of Fed. Spec. QQ-A-250/4. As an alternate, the frame shall be fabricated of 6061 aluminum alloy, conforming to temper T6 of Fed. Spec. QQ-A-250/11.

3.2.2 Scale Support. The scale support shall be fabricated of materials as specified in 3.2.1, etched aluminum alloy, commercially designated 5052-H32 or 5052-H34, in accordance with Fed. Spec. QQ-A-250/8. suitable for the purpose intended.

3.2.3 Pointer. The pointer shall be fabricated of spring steel or harddrawn stainless steel conforming to Class 3 material, of Fed. Spec. GG-I-526 except hardness and salt spray test requirements shall not apply.

* 3.2.4 Scale. The scale shall be printed on a plastic strip.

* 3.2.5 Remainder of Mechanical Tonometer. The remainder of the mechanical tonometer (including weights and test block) shall conform to Class 3 of Fed. Spec. GG-I-526, except hardness and salt spray test requirements shall not apply. As an alternate, the hammer, handle-arms and handle-sleeve shall be made of bright chrome plated brass material.

3.2.6 Case. The case for the mechanical tonometer shall be constructed from a high impact resistant material. There shall be a pressure pad or a form fitting molded insert in the case to accommodate the unit and weights to prevent their shifting and damage during transit or storage.

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3.3 Style, design and dimensions. The mechanical tonometer shall be Schiotz design as approved by the Committee on Standardization of the American Academy of Ophthalmology and Otolaryngology. The instrument shall consist of a rigid housing having a limited vertical motion within the retaining handle. There shall be a freely-moving plunger within the housing. This plunger shall carry weight or weights as described hereinafter, and shall transmit its vertical motion to a pointer by means of a lever. The motion so transmitted shall be measured by means of a scale located at the top of the instrument and over which the pointer shall pass. The scale shall be graduated from at least -1 to at least + 15 in one unit steps.

* 3.3.1 Weights. Four weights shall be supplied with the instrument. One weight marked "5.5" shall be secured to the plunger by clipping action. The other three weights shall be furnished: one 2-gm. weight marked marked "7.5", one 4.5-gm. weight marked "10", and one 9.5-gm. weight marked "15". The weights shall each have a tolerance of ± 0.02 gm. These weights shall be put on or off easily yet they shall be secured to the shaft.

3.3.2 Dimensions. Dimensions shall be in accordance with the specification 5 or latest edition for tonometer, as published by the Committee on Standardization of Tonometers of the American Academy of Opthalomology and Otolaryngology. Where dimensional requirements contained in this specification are in conflict with the committee's latest specification, the latter specification shall govern. Dimensions are given in millimeters (mm.).

3.3.2.1 Footplate

Diameter	10.1	± 0.2
Radius of concavity	15.0	± 0.25
Diameter of footplate hole	3.7	±0.5

3.3.2.2 Plunger

Diameter	3.0	± 0.03
Radius of concavity of base	15.0	± 0.75
Extension of plunger below spherical surface of footplate	3 maximum	

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3.3.2.3 Pointer

Depth	3 maximum
Thickness	0.25 maximum
Distance between scale and pointer	1.0 maximum

3.3.2.4. Hammer

Plunger contact with hammer	5 to 10

3.3.3 Scale. Each scale unit shall equal 0.050 mm. of plunger displacement. The tonometer scale shall extend from -1.00 at least + 15.0. A test block shall be supplied in the case with each tonometer. The tonometer when resisting on the convexity of the test block shall give a reading of 0.00 ± 0.15. The scale shall be calibrated so that the plunger displacement in mm. from -1.0 scale reading shall be as follows:

Scale Reading	Plunger Displacement
5.0	0.30 ± 0.01
10.0	0.55 ± 0.015
15.0	0.80 ± 0.03

3.4 Finish. The finish shall be mirror, or mirror and satin, except the pointer, which shall be given a blue or black finish. As an alternate, frame shall be fine sandblasted and anodized. Finish shall be such that it will not chip.

* 3.5 Chart. A chart converting scale units to mm. of mercury shall be furnished with each instrument in printed form.

3.6 Case. Each instrument shall be provided with a lined case suitably fitted to securely retain the instrument and accessories (see 3.2.5). Outside dimensions of the case shall not exceed 6-1/2 X 4-1/4 X 1-3/16 inches.

3.7 Identification marking. Each instrument shall be permanently and legibly marked with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.8 Instructions for Disinfecting and Cleaning Mechanical Tonometers.
The following instructions for disinfecting and cleaning mechanical tonometers as approved by the Committee on Standardization of Tonometers of the American Academy of Ophthalmology and Otolaryngology shall be furnished and packed into each instrument case:

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Instructions for Disinfection and Cleaning Mechanical Tonometers

(1) Disinfection.

In order to avoid the transference of infectious bacteria or viruses from one patient to another, the end of the mechanical tonometer coming in contact with the eye shall be placed in suitable stand containing a 1-5000 Benzalkonium Chloride Solution (U.S.P.) so that the mechnical tonometer footplate and plunger remain immersed in this disinfecting solution during the period of day in which the tonometer is in use. The instrument shall not be left in the disinfecting solution overnight. Since the disinfectant may be slightly irritating to the eye, the footplate and the lower end of the plunger shall be washed with distilled water and wiped dry with a gauze sponge immediately before use. If a suitable stand is not available, the mechanical tonometer footplate and lower end of the plunger shall be wiped with a gauze sponge moistened with 1-5000 Benzalkonium Chloride Solution (U.S.P), rinsed with distilled water, and wiped dry with a gauze sponge immediately before and after each use.

(2) Cleaning.

At the end of the day's use, the plunger shall be removed from the cylinder of the mechanical tonometer after removing the 5.5gram weight. The plunger shall be wiped clean with a gauze sponge moistened with 70 to 95 percent alcohol. The plunger casing of the cylinder shall be thoroughly irrigated with warm distilled water to rapidly dissolve any crystals of salt which may have formed from tear fluid. The plunger casing shall then be irrigated with 70 to 95 percent alcohol, and the mechanical tonometer allowed to dry. The mechanical tonometer and the plunger shall then be kept in its closed carrying case until the next day's use.

3.9 Workmanship. Mechanical tonometers shall be free from defects which detract from its appearance or impair its serviceability. There shall be no sharp or rough metal edges or corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used herein, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificate of quality. When available, certificates of quality, supplied by the manufacturer of the component or material listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Unless otherwise specified, at least 2 tonometers shall be taken at random from each 100 or fraction thereof delivered.

4.2.1 Sampling unit usage. Where possible the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Testing. Tests shall be performed in compliance with the requirements set forth in the latest specification of the Committee on Standardization of Tonometers of the American Academy of Ophthalmology and Otolaryngology.

4.4 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. Packing

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Components comprising each tonometer shall be positioned in a carrying case as specified in 3.2.6. The case shall be further packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.1.2 Intermediate package. Six unit packages shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the Box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for tonometers shall conform to the requirements of MIL-STD-794 for level C.

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5.1.2.2 Intermediate package. Six unit packages shall be packaged in accordance with the requirements of lMIL-STD-794 for level C.

5.2 Packing. Packing shall be labeled A, B, or C as specified (see 6.1).

5.2.1 Level A. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. Eight intermediate packages (48 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5. Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantity less than 40 cubic feet neeed not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Level A and B. Tonometers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Tonometers packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

* 5.3 Marking.

5.3.1 Civil agencies. Marking shall be as specified in FED-STD-123 and shall include any special markings specified in the contract or order.

5.3.2 Military.

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5.3.2.1 Levels A, B and C. Each unit package, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

* 5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).
d. Identification marking required (see 5.3).
e. Sampling required (see 4.2).

6.2 This specification is intended to cover the type normally procured to meet civil agency and military requirements. It does not include all types, as indicated by the title nor all types which are commercially available.

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-T-585B	TONOMETER, OPHTHALMIC, SCHIOTZ

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
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